GO2 Quadruped Tutorials

This package supplies Sphinx-based tutorial content to assist you with setting up and operating your GO2 quadruped robot. The tutorials topics are listed in the left column, and presented in the suggested reading order.

Important

Please be aware that the large language model and login information when connected over the internet has the **information transferred** to Unitree throughout.

The **Unitree Go2** represents the new frontier of *embodied AI* — a powerful fusion of robotics and intelligence that redefines the future of bionic quadrupeds. Designed with state-of-the-art perception, advanced AI-driven locomotion, and robust mechanical performance, the Go2 is engineered to explore, learn, and interact with the world around it.



Unitree's relentless innovation and technological leadership have led to the creation of the Go2 — an **intelligent, agile, and reliable partner** for research, exploration, and real-world applications.

Revolutionary Perception with 4D LiDAR

The Go2 introduces the **Unitree self-developed 4D LiDAR L1**, offering a **200% improvement** in recognition performance. With an ultra-wide hemispherical field of view (360° × 90°), a *minimum detection distance of just 0.05m*, and exceptionally small blind spots, the Go2 is capable of **true all-terrain recognition** and safe navigation.

Embodied Intelligence: AI + Robot

Powered by **GPT-driven intelligence**, the Go2 integrates *AI with robotics* to achieve a new level of **decision-making and adaptability**. Through **large-scale AI simulation training**, the robot masters complex gaits including:

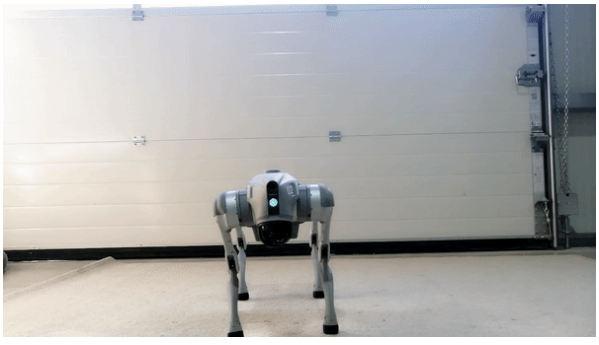
- Upside-down walking
- Adaptive roll-over recovery
- Obstacle climbing
- Agile, stable locomotion in dynamic environments

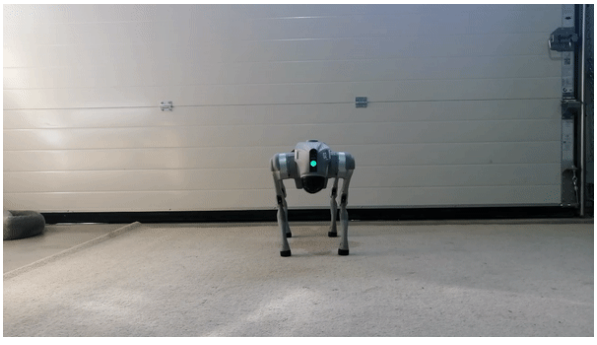
This makes the Go2 one of the most **versatile and intelligent quadrupeds** in its class.

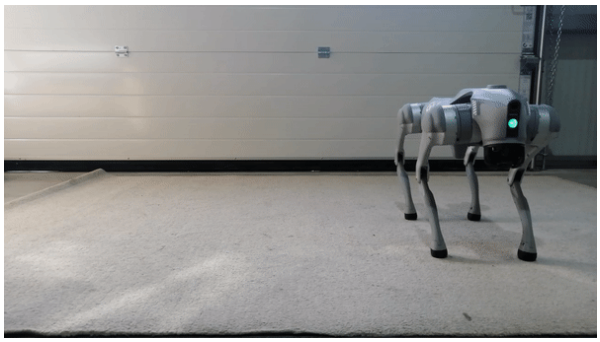
Dynamic Movements and Expressive Poses

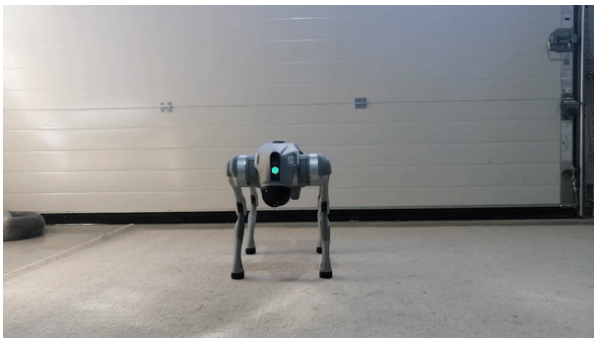
The Go2 is designed not only for **functional mobility** but also for **expressive interaction**. It can perform a wide variety of actions and poses, such as:

- Jumping and stretching
- Shaking hands
- Cheering and pouncing
- Sitting and resting









Enhanced Battery and Power System

With an upgraded **8,000mAh battery** (standard) and an optional **15,000mAh ultra-long-life battery**, the Go2 achieves **up to 150% greater endurance** compared to previous models. Its **28.8V system voltage** enhances *motor efficiency, stability, and overall performance*.

Superior Motor Performance

The Go2's redesigned actuators deliver **30% improved performance** with a **peak joint torque of 45 N·m**. A new *internal trace connection technique* and integrated *heat pipe cooling* ensure stability during long-term operation, even under demanding workloads.

3D LiDAR Mapping and Autonomy

Using the built-in **4D LiDAR L1** and a dedicated **mobile App**, users can create **3D point cloud maps** and define navigation paths. This enables the Go2 to **move autonomously** and adapt intelligently to complex environments.

ISS 2.0 — Intelligent Side-Follow System

The Go2 is equipped with the latest **ISS 2.0 (Intelligent Side-follow System)**. With advanced *wireless vector positioning and control*, it offers:

- **50% improvement in positioning accuracy**
- **Remote control distance >30m**
- **Optimized obstacle avoidance strategies**

This ensures smooth and intelligent traversal over challenging terrains.

OTA — Continuous Evolution

The Go2 is designed to **keep evolving**. With *user authorization*, it connects to the **cloud-based OTA service**, allowing seamless program upgrades that continuously improve performance and expand functionality.

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Technical Specifications

Core Capabilities	
Recognition System	Standard Ultra-Wide 4D LiDAR L1
Maximum Running Speed	~5 m/s (tested in lab conditions)
Peak Joint Torque	~45 N·m
Wireless Modules	Wi-Fi 6 / Bluetooth / 4G
Battery Endurance	~2–4 hours (depending on configuration and use)
Side-follow System	ISS 2.0

Intelligent Interaction

- **Real-time HD image transmission** via the mobile App
- **4G & eSIM support** for stable long-range connectivity
- **Graphical programming interface** with drag-and-drop functions for beginners and developers
- **Advanced obstacle avoidance** using 4D LiDAR with precise 3D world reconstruction

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Pushing Boundaries with Technology

The **Unitree Go2** is more than just a quadruped robot — it is a **new intelligent creature**. With advanced AI, superior hardware, and a constantly evolving system, the Go2 leads the **next generation of bionic robotics**, setting a new standard for performance, intelligence, and interaction.

Important

- The listed specifications may vary based on configuration and usage environment.
- Maximum torque values differ between joints; 45 N·m represents peak performance at the largest joint.
- Battery endurance depends on operational conditions and selected battery configuration.
- Product features may continue to evolve via OTA updates.
- Always refer to the actual product for the latest details.

- This product is intended for civilian applications only; avoid unsafe modifications or hazardous usage.

Model Variants ›

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